

Homework 3 - Stat 495 - Fall 2025

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2025-10-08

Homework 3 - Due Wednesday, Oct. 8th by midnight to Gradescope

Practicing Academic Integrity

If you worked with others or used resources outside of provided course material, such as Generative AI (anything besides our textbook, course materials in the repo, labs, R help menu) to help complete this assignment, please acknowledge them below using a bulleted list. Remember that according to our class framework, if you engage with Generative AI tools, the tool and date of use must be listed here, and then a complete record of the chat history must be appended to the assignment.

I acknowledge the following individuals with whom I worked on this assignment:

Name(s) and how they assisted you

-

I used the following sources to help complete this assignment:

Source(s) and where used in your submission

-

PROBLEMS TO TURN IN: Add 1, Add 2, and Add 3

Additional 1 - Ship Damage

```
ships <- MASS::ships  
# ?ships # for reference if you want more information  
names(ships)
```

```
[1] "type"      "year"      "period"    "service"   "incidents"
```

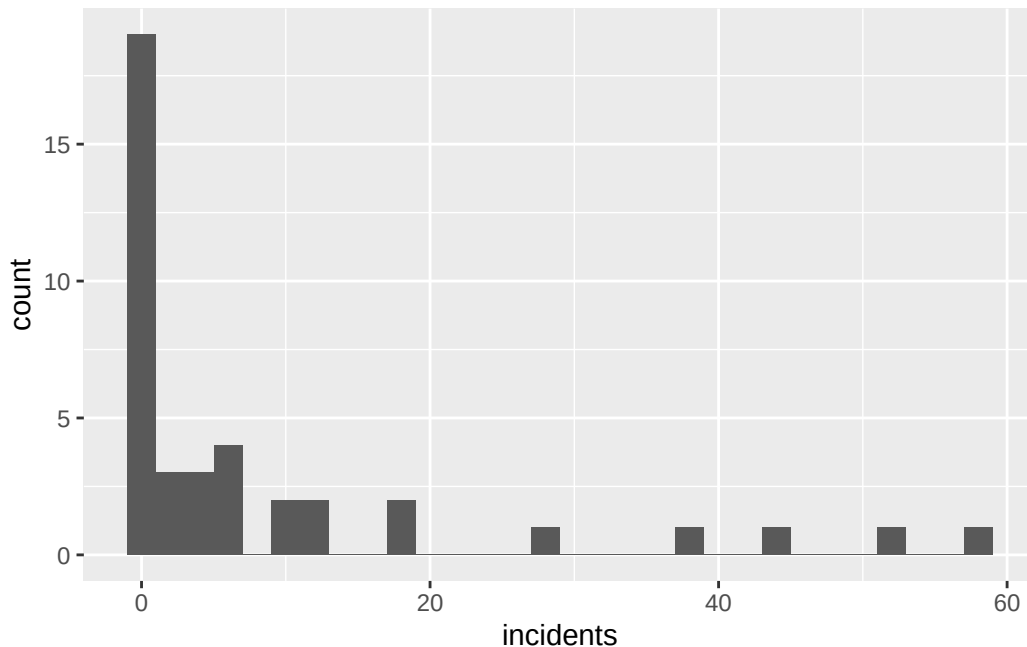
This data set contains information on ships and the number of damage incidents they have suffered along with a few potential predictors. We want to predict the number of damage incidents using other predictors. Note that due to the small data set size, we will not be using a training/test split or CV for analysis. Instead, just fit your models to the full data set to address the questions below.

part a: Plot the distribution of the response variable, incidents. Explain why the usual OLS approach should not be used here and what you should do instead, using the plot as support.

SOLUTION: The usual OLS should not be used because the response variable is discrete (not continuous) and from the histogram we see that the normality assumption required by OLS is violated. We should use Poisson regression instead since the response variable is a count.

```
ggplot(data = ships, aes(x = incidents)) +  
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



part b: Fit an appropriate regression predicting *incidents* using *service*. Obtain the model summary. Is the overall model significant? (Use a likelihood ratio test through the package *lmttest* to do the drop-in deviance procedure.)

SOLUTION: Yes, the overall model is significant as evidenced by the very small p-value ($<2e-16$) which shows that **service** is a statistically significant predictor of **incidents**.

```
poismod <- glm(incidents ~ service , data = ships,
               family = poisson (link = log))
msummary(poismod)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.61e+00	7.15e-02	22.6	<2e-16 ***
service	6.42e-05	2.87e-06	22.4	<2e-16 ***

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 730.25 on 39 degrees of freedom
 Residual deviance: 374.55 on 38 degrees of freedom
 AIC: 476.4

Number of Fisher Scoring iterations: 6

```
lrtest(poismod)
```

Likelihood ratio test

Model 1: incidents ~ service

Model 2: incidents ~ 1

```
#Df LogLik Df Chisq Pr(>Chisq)
1    2 -236.2
2    1 -414.1 -1 355.7    <2e-16 ***
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

part c: Fit an appropriate regression predicting *incidents* using both *service* and *type* including their interaction. Obtain the model summary. Compare this model to the one with just *service*. Do you need to keep the interaction? Do you need to keep *type*? Perform appropriate procedures to assess (you may need to fit other models to answer the questions).

SOLUTION: A likelihood ratio test comparing the model with interaction (*poismod2*) to the simpler model with only *service* (*poismod*) shows that *poismod2* is overall more statistically significant, as evidenced by the very small p-value (<2e-16). Also, *poismod2* has a substantially lower AIC value compared to *poismod* (267.2 vs. 476.4). This indicates that having *type* as a predictor in addition to service makes the model perform better.

To assess whether the interaction is necessary, *poismod2* is compared with a third model that includes *service* and *type* without their interaction. The likelihood ratio test still favors *poismod2* as evidenced by the very small p-value (<2e-16). *poismod2* also has the lowest AIC value of the three models. Therefore we keep the interaction.

```
poismod2 <- glm(incidents ~ service * type, data = ships,
               family = poisson (link = log))
msummary(poismod2)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.012931	0.376646	0.03	0.97	
service	0.000911	0.000142	6.42	1.3e-10	***
typeB	2.892171	0.393861	7.34	2.1e-13	***
typeC	-0.024582	0.651251	-0.04	0.97	
typeD	-0.348387	0.592329	-0.59	0.56	
typeE	0.422829	0.496132	0.85	0.39	
service:typeB	-0.000883	0.000142	-6.23	4.8e-10	***

```
service:typeC -0.000422  0.000495  -0.85    0.39
service:typeD  0.000293  0.000329   0.89    0.37
service:typeE  0.000102  0.000255   0.40    0.69
```

(Dispersion parameter for poisson family taken to be 1)

```
Null deviance: 730.25  on 39  degrees of freedom
Residual deviance: 149.30  on 30  degrees of freedom
AIC: 267.2
```

Number of Fisher Scoring iterations: 6

```
lrtest(poismod,poismod2)
```

Likelihood ratio test

Model 1: incidents ~ service

Model 2: incidents ~ service * type

```
#Df LogLik Df Chisq Pr(>Chisq)
1    2 -236.2
2   10 -123.6  8 225.2    <2e-16 ***
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
poismod3 <- glm(incidents ~ service + type, data = ships,
                family = poisson (link = log))
msummary(poismod3)
```

Coefficients:

```
Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.62e+00  1.54e-01  10.51 < 2e-16 ***
service      2.90e-05  4.21e-06   6.89 5.5e-12 ***
typeB        1.25e+00  1.91e-01   6.55 5.9e-11 ***
typeC       -1.24e+00  3.27e-01  -3.79 0.00015 ***
typeD       -8.86e-01  2.87e-01  -3.08 0.00206 **
typeE       -2.56e-01  2.35e-01  -1.09 0.27564
```

(Dispersion parameter for poisson family taken to be 1)

```
Null deviance: 730.25  on 39  degrees of freedom
Residual deviance: 230.83  on 34  degrees of freedom
```

AIC: 340.7

Number of Fisher Scoring iterations: 6

```
lrtest(poismod3,poismod2)
```

Likelihood ratio test

Model 1: incidents ~ service + type

Model 2: incidents ~ service * type

```
#Df LogLik Df Chisq Pr(>Chisq)
1    6 -164.3
2   10 -123.6  4 81.53    <2e-16 ***
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

part d: For your preferred model (from above), save predictions from the model using *predict* on the full data set of two different types - once with type = “link” and one with type = “response”. Show the first 5 predictions of each type. (Note type here is not the ship type, but type of prediction!) Explain the difference between the prediction types, how you can map back and forth between them, and why that is the case.

SOLUTION: The “link” prediction type returns predicted values on the link scale, which in this case corresponds to the log of the expected count of **incidents**. The “response” prediction type returns predictions on the response scale, which means the expected count of **incidents** itself. To map between the two prediction types, exponentiate the link predictions to get response predictions and take the log of the response predictions to return to the link scale. This is because in a Poisson regression, the link function is the log.

```
aug_link <- augment(poismod2, type.predict = "link")
```

```
aug_resp <- augment(poismod2, type.predict = "response")
```

```
head(aug_link$.fitted, 5)
```

```
[1] 0.128571 0.070296 1.009983 1.009983 1.389683
```

```
head(aug_resp$.fitted, 5)
```

```
[1] 1.13720 1.07283 2.74556 2.74556 4.01358
```

Additional 2 - Soil predictions

We used this data set before, and will try new prediction methods on it here.

The data comes from a Kaggle competition: <https://www.kaggle.com/c/afsis-soil-properties>. The original data set contained 3600 variables, 3599 possible predictors (really, 3578 and some other variables) and a response, Sand. The 3599 predictors were reduced to 106 that can “best” distinguish between the two levels of Depth (another variable in the data set). The resulting data set of 107 variables (106 predictors and the response variable, Sand) was saved in the data set “newsoil”. The row numbers should be removed as demonstrated below. We want to use this data set due to the large number of predictors to see what our shrinkage methods do.

```
newsoil <- read.csv("https://awagaman.people.amherst.edu/stat495/newsoil.csv",
                    header = T)
newsoil <- select(newsoil, -X)
```

Our focus is on predicting the response variable Sand, using the selected variables from previous work.

Trees are a prediction method that have an inherent variable selection mechanism. Not all predictors get used in the tree! In this way, you could imagine comparing the solution here to automated variable selection and lasso solutions.

For this problem, we won’t use a training/test set or CV, just so you can focus on understanding the components of a tree and its options.

part a: Considering that we want to predict the variable *Sand*, will we be constructing a classification or regression tree? Why?

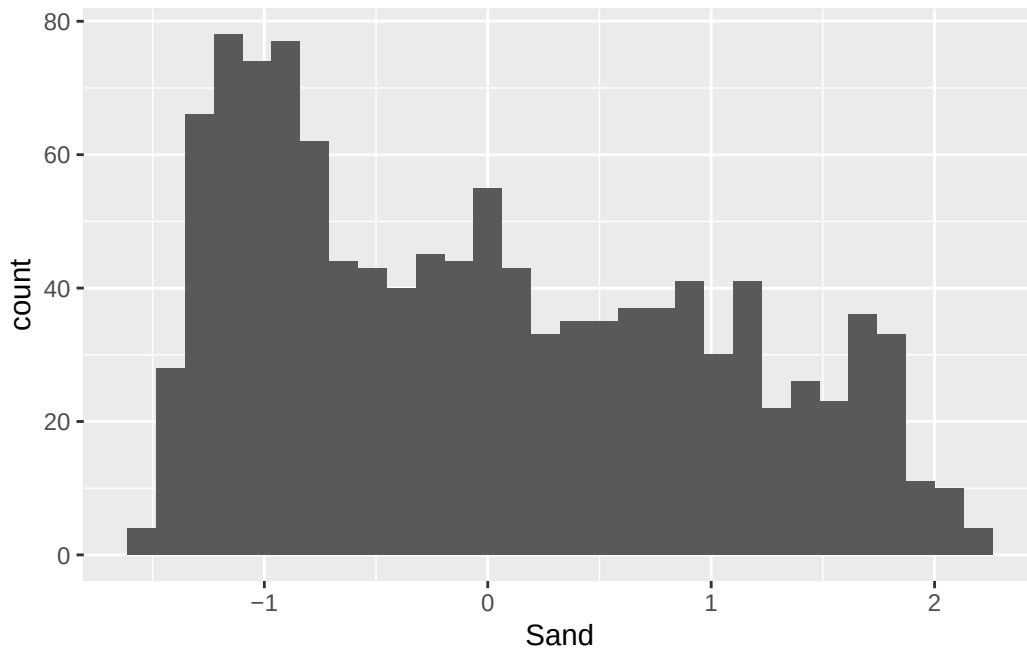
SOLUTION: We will be constructing a regression tree because **Sand** is a quantitative response variable.

```
str(newsoil$Sand)
```

```
num [1:1157] 1.27 1.69 2.09 2.12 2.16 ...
```

```
ggplot(data = newsoil, aes(x = Sand)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



part b: Fit an appropriate tree with no control options set (default tree) using *rpart*, obtain the summary using *printcp*, and plot the resulting tree. How many variables were used in the tree construction?

SOLUTION: 11 variables were used in the tree construction.

```
sand.rpart <- rpart(Sand ~ . , data = newsoil, method = "anova")
printcp(sand.rpart)
```

Regression tree:

```
rpart(formula = Sand ~ . , data = newsoil, method = "anova")
```

Variables actually used in tree construction:

```
[1] m1826.28 m1835.92 m1837.85 m1839.78 m3600.49 m3619.77 m3669.91 m3685.34
[9] m3710.41 m3741.27 m3748.98
```

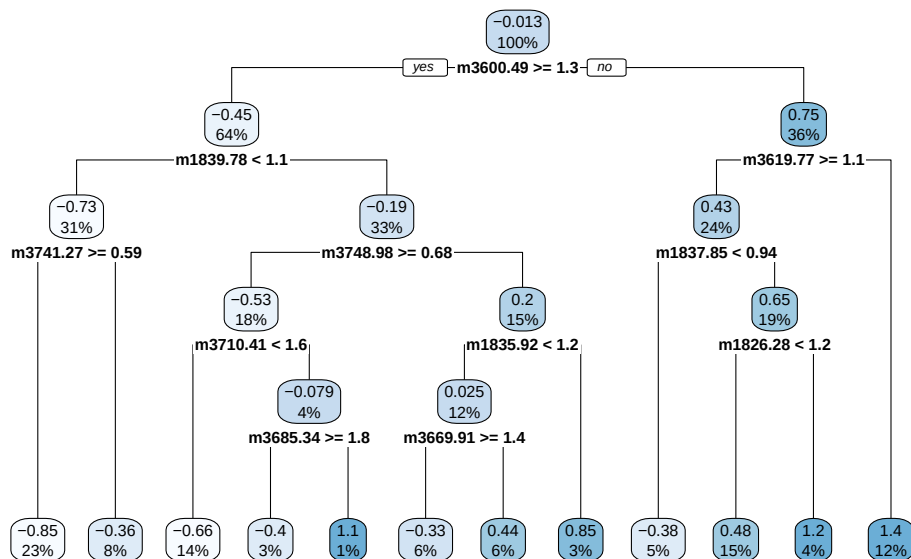
Root node error: $1130/1157 = 0.9763$

n= 1157

	CP	nsplit	rel error	xerror	xstd
1	0.33913	0	1.0000	1.0046	0.02898

2	0.07462	1	0.6609	0.6730	0.02841
3	0.04635	2	0.5863	0.6054	0.02758
4	0.04561	3	0.5399	0.5571	0.02636
5	0.04303	4	0.4943	0.5283	0.02546
6	0.01819	5	0.4513	0.4869	0.02470
7	0.01811	6	0.4331	0.4777	0.02538
8	0.01807	7	0.4150	0.4772	0.02533
9	0.01381	8	0.3969	0.4577	0.02535
10	0.01263	9	0.3831	0.4431	0.02480
11	0.01000	11	0.3578	0.4401	0.02480

```
rpart.plot(sand.rpart)
```



part c: Fit a second tree with control options set as follows: `control = c(cp=0, minsplitlevel = 20)`. Obtain the summary using `printcp`. (Plotting the tree would show something you can't really read, so don't worry about that.) How many variables were used in the tree construction?

SOLUTION: 82 variables were used in the tree construction.

```
sand.rpart2 <- rpart(Sand ~ . , data = newsoil,
                     control = rpart.control(cp = 0, minsplitlevel = 20))
printcp(sand.rpart2)
```

Regression tree:

```
rpart(formula = Sand ~ ., data = newsoil, control = rpart.control(cp = 0,
  minsplit = 20))
```

Variables actually used in tree construction:

```
[1] m1546.65 m1552.43 m1556.29 m1560.15 m1564      m1569.79 m1571.72 m1575.58
[9] m1577.5  m1820.49 m1822.42 m1824.35 m1826.28 m1835.92 m1837.85 m1839.78
[17] m3600.49 m3606.27 m3612.06 m3613.99 m3615.92 m3617.84 m3619.77 m3621.7
[25] m3627.49 m3629.42 m3646.77 m3648.7  m3652.56 m3654.49 m3658.34 m3660.27
[33] m3662.2  m3669.91 m3679.56 m3681.48 m3683.41 m3685.34 m3687.27 m3691.13
[41] m3693.06 m3694.98 m3704.63 m3708.48 m3710.41 m3712.34 m3716.2  m3720.05
[49] m3723.91 m3725.84 m3731.63 m3733.55 m3735.48 m3741.27 m3743.2  m3745.13
[57] m3747.05 m3748.98
```

Root node error: 1130/1157 = 0.9763

n= 1157

	CP	nsplit	rel	error	xerror	xstd
1	3.391e-01	0	1.0000	1.0007	0.02886	
2	7.462e-02	1	0.6609	0.6690	0.02832	
3	4.635e-02	2	0.5863	0.6073	0.02792	
4	4.561e-02	3	0.5399	0.5629	0.02739	
5	4.303e-02	4	0.4943	0.5417	0.02670	
6	1.819e-02	5	0.4513	0.5003	0.02548	
7	1.811e-02	6	0.4331	0.4897	0.02583	
8	1.807e-02	7	0.4150	0.4897	0.02583	
9	1.381e-02	8	0.3969	0.4756	0.02569	
10	1.263e-02	9	0.3831	0.4693	0.02595	
11	9.583e-03	11	0.3578	0.4492	0.02525	
12	8.928e-03	12	0.3482	0.4492	0.02522	
13	7.886e-03	13	0.3393	0.4507	0.02549	
14	7.228e-03	14	0.3314	0.4613	0.02587	
15	7.108e-03	15	0.3242	0.4693	0.02617	
16	6.981e-03	16	0.3171	0.4685	0.02613	
17	6.522e-03	17	0.3101	0.4641	0.02616	
18	5.813e-03	18	0.3036	0.4643	0.02625	
19	5.345e-03	19	0.2978	0.4657	0.02664	
20	4.882e-03	20	0.2924	0.4608	0.02648	
21	4.535e-03	22	0.2827	0.4619	0.02698	
22	4.528e-03	23	0.2781	0.4642	0.02718	
23	4.247e-03	26	0.2646	0.4678	0.02775	

24	4.231e-03	27	0.2603	0.4643	0.02776
25	3.996e-03	28	0.2561	0.4638	0.02794
26	3.823e-03	29	0.2521	0.4594	0.02798
27	3.600e-03	30	0.2483	0.4633	0.02825
28	3.398e-03	31	0.2447	0.4652	0.02833
29	3.266e-03	33	0.2379	0.4638	0.02835
30	3.137e-03	34	0.2346	0.4584	0.02825
31	3.054e-03	35	0.2315	0.4560	0.02811
32	3.003e-03	36	0.2284	0.4565	0.02811
33	2.736e-03	37	0.2254	0.4546	0.02814
34	2.464e-03	38	0.2227	0.4556	0.02826
35	2.068e-03	39	0.2202	0.4603	0.02865
36	2.026e-03	40	0.2181	0.4647	0.02905
37	1.894e-03	41	0.2161	0.4643	0.02914
38	1.852e-03	43	0.2123	0.4629	0.02914
39	1.842e-03	44	0.2105	0.4629	0.02914
40	1.832e-03	45	0.2086	0.4633	0.02914
41	1.818e-03	46	0.2068	0.4632	0.02914
42	1.788e-03	47	0.2050	0.4613	0.02899
43	1.732e-03	51	0.1978	0.4607	0.02897
44	1.723e-03	52	0.1961	0.4617	0.02899
45	1.687e-03	53	0.1944	0.4629	0.02908
46	1.505e-03	56	0.1893	0.4600	0.02910
47	1.491e-03	57	0.1878	0.4622	0.02952
48	1.476e-03	59	0.1848	0.4628	0.02952
49	1.409e-03	60	0.1833	0.4638	0.02959
50	1.224e-03	61	0.1819	0.4651	0.02963
51	1.220e-03	62	0.1807	0.4664	0.02965
52	1.198e-03	63	0.1795	0.4664	0.02965
53	1.192e-03	64	0.1783	0.4675	0.02966
54	1.080e-03	65	0.1771	0.4676	0.02967
55	9.717e-04	66	0.1760	0.4707	0.02996
56	9.098e-04	67	0.1750	0.4715	0.03002
57	9.082e-04	68	0.1741	0.4719	0.03005
58	8.747e-04	70	0.1723	0.4727	0.03005
59	8.492e-04	71	0.1714	0.4735	0.03005
60	8.040e-04	72	0.1706	0.4728	0.03005
61	7.604e-04	73	0.1698	0.4726	0.03005
62	7.411e-04	74	0.1690	0.4733	0.03008
63	7.340e-04	76	0.1676	0.4734	0.03008
64	7.275e-04	79	0.1652	0.4733	0.03008
65	7.208e-04	80	0.1645	0.4743	0.03028
66	5.969e-04	81	0.1638	0.4761	0.03036

67	5.883e-04	82	0.1632	0.4777	0.03013
68	5.056e-04	83	0.1626	0.4776	0.03014
69	4.705e-04	84	0.1621	0.4784	0.03014
70	4.551e-04	85	0.1616	0.4777	0.03014
71	4.197e-04	86	0.1612	0.4782	0.03014
72	4.170e-04	87	0.1607	0.4788	0.03015
73	2.593e-04	89	0.1599	0.4787	0.03016
74	2.448e-04	90	0.1596	0.4780	0.03013
75	1.595e-04	91	0.1594	0.4781	0.03014
76	1.449e-04	92	0.1592	0.4772	0.02979
77	1.273e-04	94	0.1590	0.4772	0.02979
78	1.082e-04	95	0.1588	0.4770	0.02977
79	1.019e-04	96	0.1587	0.4770	0.02977
80	8.577e-05	97	0.1586	0.4772	0.02978
81	5.191e-05	98	0.1585	0.4770	0.02977
82	0.000e+00	100	0.1584	0.4772	0.02978

part d: The difference between the trees in parts b and c is a change in the control parameters. The default minsplit is 20, so that actually didn't change between them. What did change between them? What was the impact in the tree generated? Explain.

SOLUTION: The complexity parameter (cp) reduced (changed) and this caused the number of splits to increase thus generating a larger and more complex tree. This is because a smaller cp reduces the minimum improvement in model fit required for a split to be made, and this allows more splits to occur since more of them are able to meet the threshold value defined by the cp.

part e: Explain how to use the tree from part b (smaller tree) to obtain a prediction for observation 495, walking through the tree splits until the prediction is reached.

SOLUTION: Observation 495 follows the path: at node m3600.49 $\geq 1.3 \rightarrow$ yes because $1.64954 > 1.3$, at node m1839.78 $< 1.1 \rightarrow$ yes because $1.0622 < 1.1$, at node m3741.27 $\geq 0.59 \rightarrow$ yes because $0.76611 > 0.59$. This then leads to a leaf node predicting -0.85.

```
newsoil |>
  select(m1826.28, m1835.92, m1837.85, m1839.78, m3600.49, m3619.77, m3669.91, m3685.34, m3710.41, m3741.27, m3748.98)
  slice(495)
```

	m1826.28	m1835.92	m1837.85	m1839.78	m3600.49	m3619.77	m3669.91	m3685.34
1	1.07582	1.06566	1.06423	1.0622	1.64954	1.77353	1.609	1.54446
	m3710.41	m3741.27	m3748.98					
1	1.34637	0.76611	0.720013					

part f: What is the residual for observation 495 using the tree from part b?

SOLUTION: The residual is -0.289727.

```
newsoil |>
  select(Sand) |>
  slice(495)
```

```
      Sand
1 -1.13562
```

```
fitted <- predict(sand.rpart)
residual <- -1.13562 - fitted[495]
residual
```

```
      495
-0.289727
```

Additional 3 - Tailored Project Reflection

Think about your recently submitted tailored project. Write a reflection (more than a paragraph but doesn't need to be more than a page) that addresses the following questions:

- Summarize your accomplishments. Did you accomplish what you set out to do? What, if anything, did you adjust along the way?
- Does the project contribute to your portfolio in the way that you hoped? Explain.
- What, if anything, was left undone, and why?
- What would you want to do differently next time (can be specific to this project or thinking broadly about managing a project in general) and why?
- What are you most proud of in regard to the project? Why?

I believe I accomplished what I set out to do. I initially planned to build the interactive dashboard using Power BI, but partway through the project I discovered that sharing Power BI dashboards publicly would be difficult without institutional access. This led me to pivot to Tableau, which required extra time to learn and adapt my workflow. Although this change delayed my progress, it ultimately became one of the most valuable parts of the experience. This project significantly enhanced my portfolio because I learned how to use another industry-standard tool to communicate business insights effectively to both technical and non-technical stakeholders. Now that I have the foundational skills, I plan to create more dashboards in the near future to continue improving my understanding and creativity in designing interactive data visualizations.

Next time, I will start by confirming that I can share my project publicly before beginning. In this project, I realized late that I couldn't share Power BI dashboards, and my other deliverable, the HTML blog site, couldn't be hosted on GitHub because our class repositories are private. (I provided alternate ways to access it on a README document, and I am also happy to walk through the process in person) I also plan to improve my commit habits in future projects. Since I often work individually for projects, I haven't always seen the need to push changes frequently to GitHub, even though I consistently save my work locally. However, I'll be more intentional about version control moving forward.

I'm proud that I was able to complete the project to a level I'm pleased with, despite unexpected commitments such as job interviews that required urgent prioritization and caused me to deviate from my original timeline. Although I finished later than I had hoped, I'm grateful that I was able to complete the project. Overall, this experience reaffirmed my passion for business intelligence.